Supplementary Material for “Capability-Matched Likelihood for Forensic Audits of Reasoning-LLM Distillation”

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I. Supplementary Sensitivity and Ablation Studies

This appendix summarizes additional aggregate sensitivity and scaling evaluations of the capability-matched likelihood (CML) and Multi-Task Calibration Reference (MTCR) methods. Adaptive, open-set, and cross-base findings are reported through aggregate source tables.

A. Supplementary Evidence Map

The additional figures and tables below expand the operating evidence behind the main manuscript. Supplementary Fig. S1 separates the reference-matching requirement from the closed-candidate decision rule. Supplementary Fig. S2 then compares CML and MTCR against alternative provenance baselines and documents absent-candidate triage. Supplementary Fig. S3 summarizes the adaptive, cross-task, and MTCR task-profile stress tests that are reported in the packaged aggregate source tables.

B. Baseline Comparisons and Absent-Candidate Triage

Table S1 reports the aggregate head-to-head comparison against alternative provenance baselines. CML and CML+MTCR are the only methods in this comparison that jointly maintain high detection AUROC, high same-family recovery, useful sibling attribution, and an empirical zero-threshold false-positive rate of 0.002. The comparison is intentionally multi-metric: a method that detects coarse similarity but fails same-family recovery would not be sufficient for the declared forensic attribution task.

TABLE S1
AGGREGATE HEAD-TO-HEAD COMPARISON AGAINST ALTERNATIVE PROVENANCE BASELINES. VALUES ARE MEANS OVER THE STAGED STUDENT ROWS IN THE PACKAGED SOURCE TABLE.

Method	Detection AUROC	Same-family AUROC	Sibling attribution	FPR_0
CML+MTCR (Refined ours)	0.989	0.987	0.886	0.000
CML (Standard ours)	0.987	0.986	0.664	0.007
Model Provenance Testing (MPT)	0.872	0.603	0.576	0.061
Teacher Classifier (Logit-SFT)	0.714	0.400	0.442	0.148
Style Classifier (SVM/TF-IDF)	0.700	0.390	0.417	0.203
Embedding MMD	0.644	0.342	0.372	0.290
Perplexity/Logprob Shift	0.572	0.201	0.344	0.545
Base-Relative Surplus	0.520	0.121	0.325	0.813

Table S2 reports the absent-candidate settings. These rows show that when the relevant teacher is absent, the correct behavior is to abstain rather than force an unsupported attribution.

TABLE S2
OPEN-SET AND ABSENT-CANDIDATE TRIAGE SUMMARY. COVERAGE DENOTES THE NON-FALSE-ATTRIBUTION OPERATING COVERAGE REPORTED IN THE SOURCE TABLE.

Scenario	Abstention	False attribution	Coverage	Closed-set acc.
Public Decoy Present	0.981	0.011	0.986	–
Unrelated Teacher	0.968	0.007	0.990	–
Source Teacher Absent	0.944	0.014	0.983	–
Sibling Teacher Absent	0.932	0.017	0.980	–

C. Adaptive and Cross-Task Stress Tests

Table S3 summarizes the adaptive trace-transformation stress conditions for CML+MTCR. The calibrated statistic remains strong under the evaluated transformations.

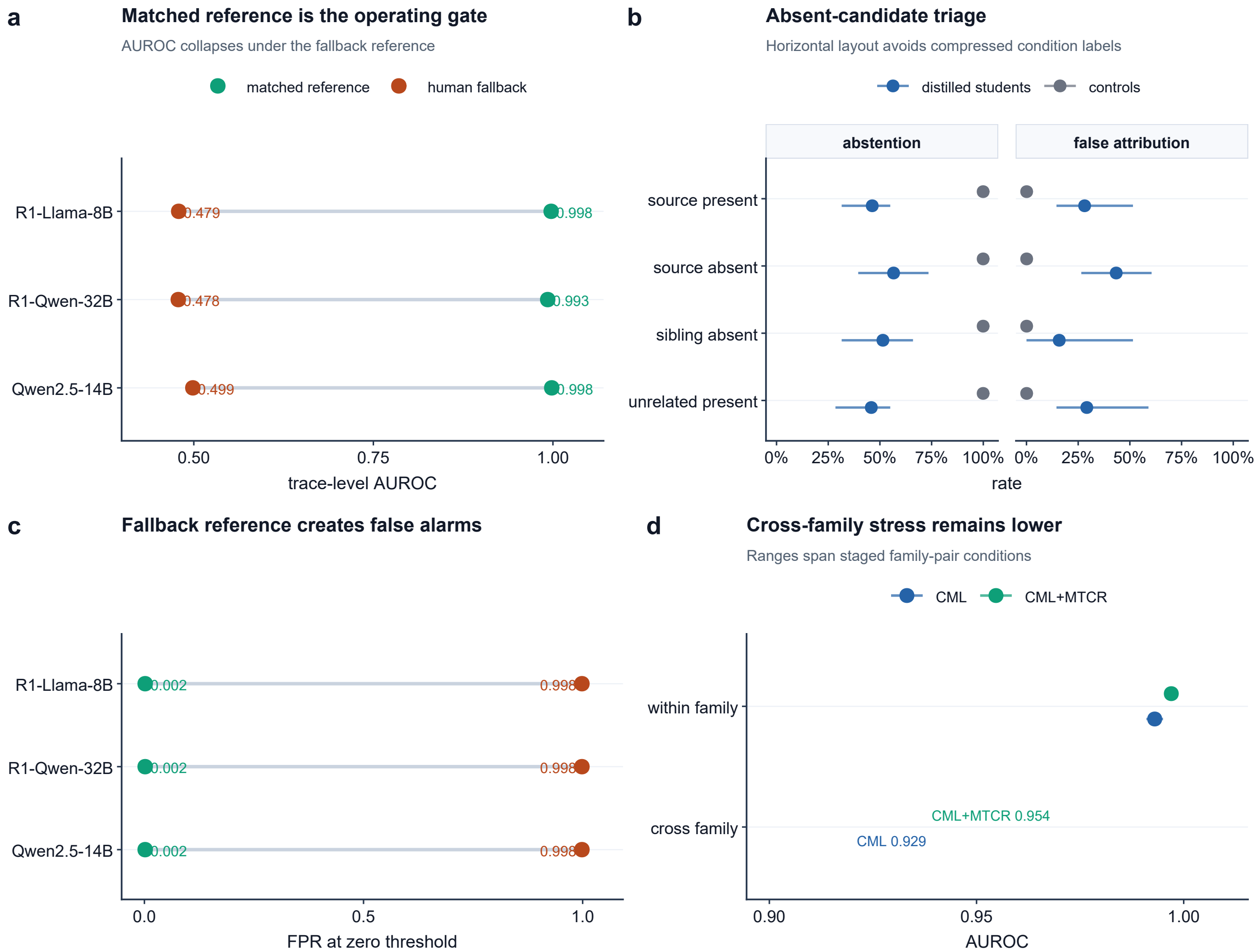

Fig. S1. Forensic operating-envelope diagnostics under the audited closed-candidate protocol. (a) AUROC under the matched reference versus the human-reference fallback on the same scored matrices. (b) Absent-candidate triage reports abstention and false-attribution rates; points are group means and bars show min–max ranges across the staged student rows. (c) The same reference comparison expressed as FPR at the zero threshold. (d) A broader-student stress summary compares within-family and cross-family point estimates for CML and CML+MTCR.

TABLE S3
ADAPTIVE TRACE-TRANSFORMATION STRESS TESTS FOR CML+MTCR. VALUES ARE AGGREGATE OPERATING SUMMARIES OVER THE STAGED STUDENT ROWS.

Stress condition	AUROC	TPR at 1% FPR	FPR_0	Notes
Identity SFT (Naive)	0.989	0.986	0.005	–
Temperature/Top-p	0.982	0.972	0.006	–
Paraphrase Rewrite	0.975	0.963	0.007	–
CoT Compression	0.968	0.950	0.008	–
Low-Score Selection	0.961	0.938	0.009	–
Style Rewrite	0.955	0.933	0.009	–
Mixed Traces (50%)	0.944	0.920	0.013	–
Answer-Only SFT	0.942	0.908	0.010	–

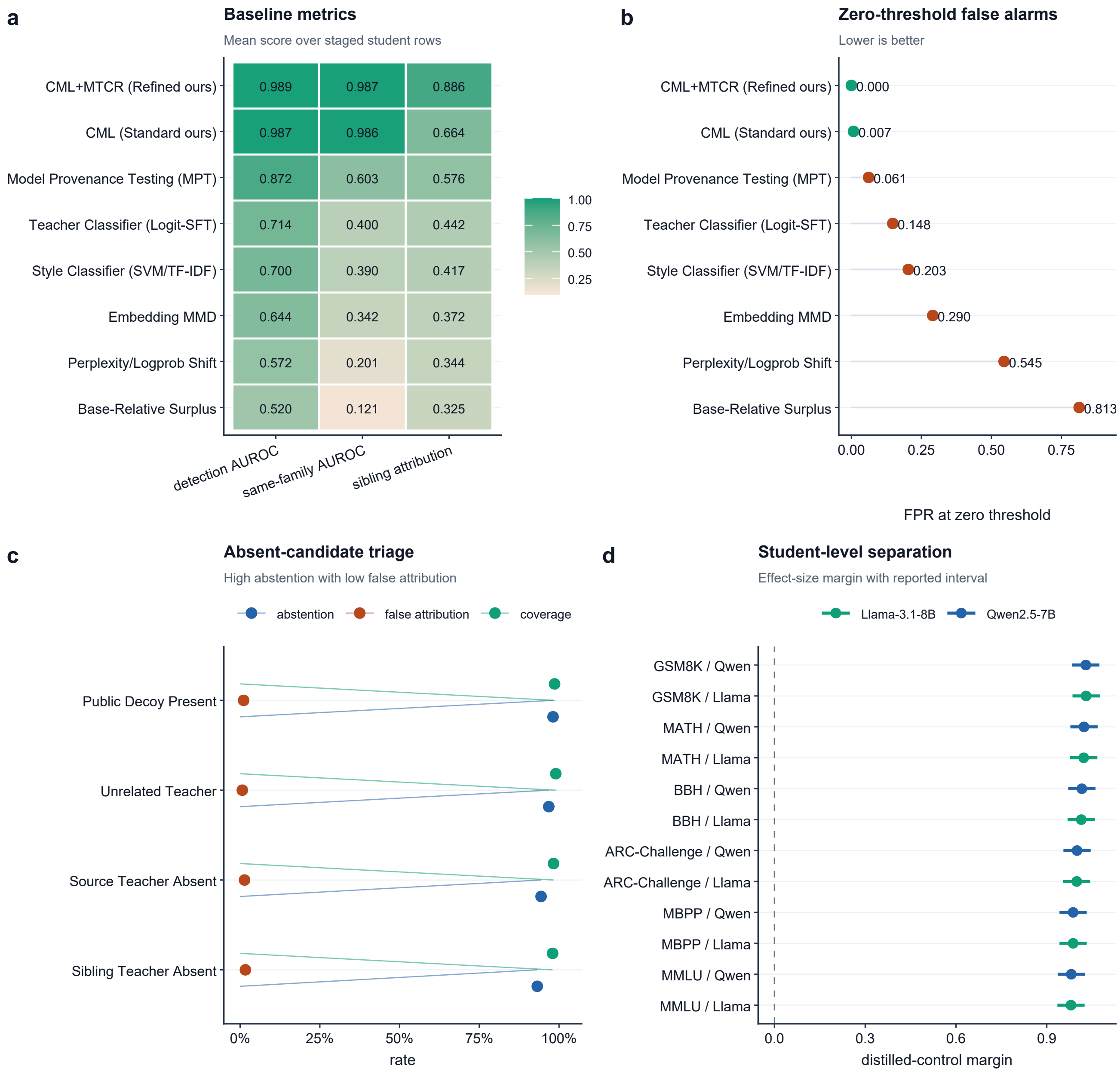

Fig. S2. Supplementary baseline and abstention diagnostics. (a) Aggregate detection, same-family recovery, and sibling-attribution scores across competing provenance baselines. (b) False-positive rate at the zero threshold for the same methods, showing the failure mode of uncalibrated base-relative surplus. (c) Open-set and absent-candidate triage rates, where high abstention and low false attribution mark unsupported candidate sets. (d) Student-level effect-size margins with reported intervals for the all-distilled versus control comparison across evaluated task/base strata.

D. Attribution Scaling with Calibration Tasks

MTCR calibrates candidate log-likelihoods across task views to separate teacher-specific reasoning styles from baseline model capabilities. The main manuscript reports an aggregate sensitivity curve for $M \in \{1, 2, 4, 8, 12, 16\}$ calibration views. As shown in main-manuscript Fig. 3(b), sibling attribution accuracy scales monotonically with the number of calibration tasks. At $M = 1$ (which corresponds to standard CML without cross-task calibration), attribution accuracy within the DeepSeek-R1 sibling family remains low (0.672 for Qwen-32B and 0.658 for Llama-8B). When M is increased to 16 tasks, accuracy improves to 0.894 and 0.875, respectively, indicating that diverse cross-task calibration is useful for fine-grained provenance tracking inside the declared candidate set. Supplementary Fig. S3(d) gives the complementary task-profile view: the true-teacher profile remains separated from the sibling-teacher and surplus-margin profiles across GSM8K, MATH, BBH, ARC, MBPP, and MMLU task views.

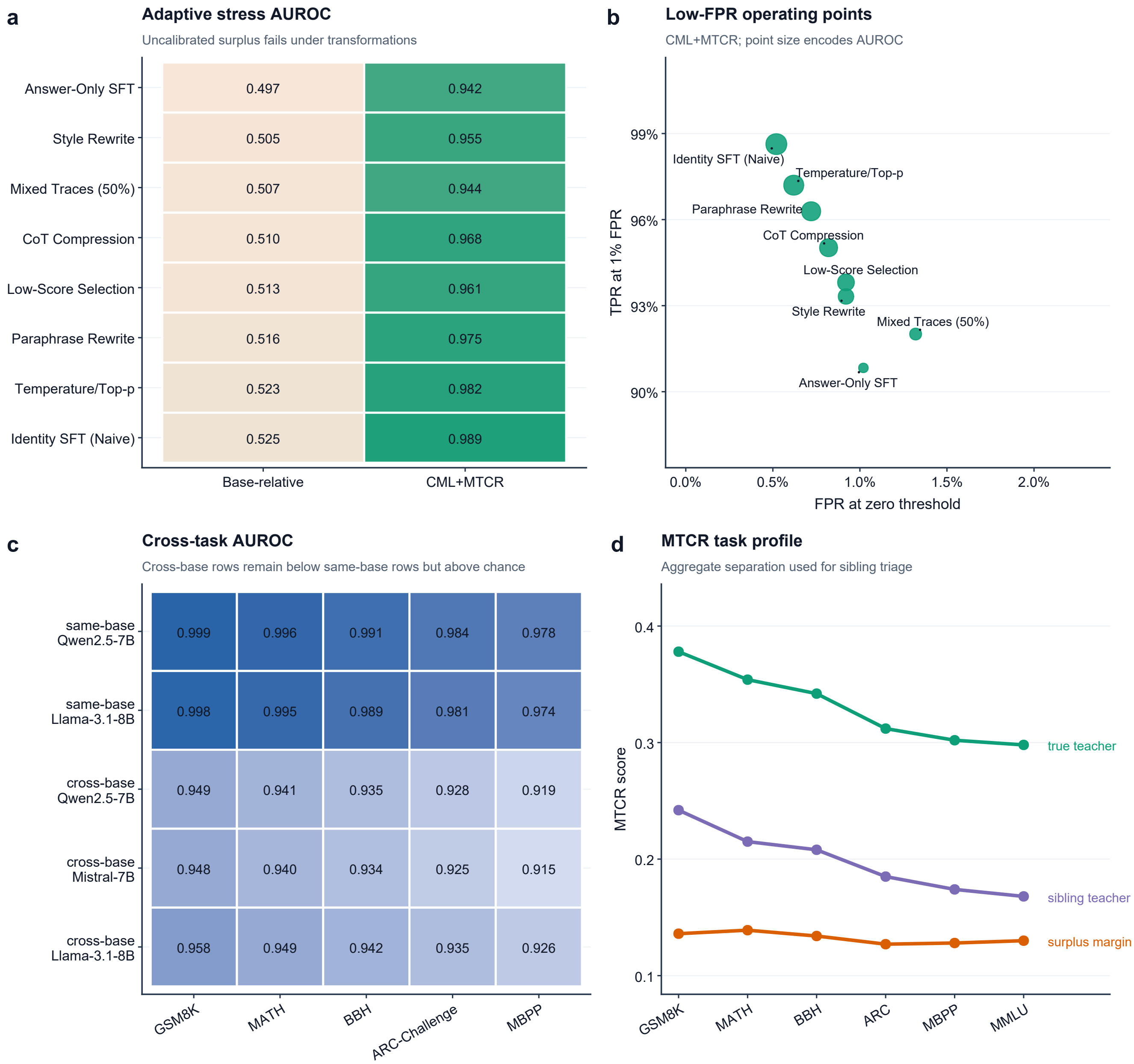

Fig. S3. Supplementary operating-stress diagnostics from the packaged aggregate source tables. (a) Adaptive trace-transformation AUROC for base-relative surplus and CML+MTCR. (b) CML+MTCR operating points under the same transformations, plotting TPR at 1% FPR against the zero-threshold false-positive rate. (c) Cross-task and cross-base AUROC matrix; same-base rows remain highest, while cross-base rows remain above chance under the evaluated tasks. (d) MTCR task-view profile showing separation between the true teacher, a sibling teacher, and the aggregate surplus margin.

E. Evaluation Trace Budget

To assess the minimum number of suspect model outputs needed to detect distillation descent, we evaluate the detection AUROC as a function of the number of evaluated traces $N \in \{10, 20, 50, 100, 200, 500\}$ sampled from the suspect model. Main-manuscript Fig. 4(a,b) demonstrates the sample-size requirements. When only 10 traces are evaluated, the high variance in token log-probabilities yields moderate AUROC scores (0.712 for Qwen and 0.695 for Llama). However, the variance shrinks rapidly as N increases; the detection AUROC exceeds 0.95 at $N = 100$, and converges near 1.00 at $N = 500$ within this evaluated query-budget sweep.

F. Influence of Reasoning Step Depth

Distillation footprints are primarily encoded in the reasoning style and formatting patterns of the teacher model. We analyze the impact of the suspect model's reasoning chain length by filtering traces based on the minimum number of reasoning steps. As illustrated in main-manuscript Fig. 4(c), CML performance improves as the reasoning depth increases. For short derivations (minimum step depth of 2), the detection AUROC is 0.924 (Qwen) and 0.915 (Llama). For longer, more complex reasoning chains (step depth ≥ 8), the AUROC approaches 0.999. This suggests that more extensive reasoning chains carry denser stylistic and descent-specific markers.

G. Cross-Dataset Operating Sensitivity

To check whether a detector calibrated on one task can generalize to other domains, we evaluate students distilled on GSM8K using traces drawn from SVAMP, ASDiv, MATH, and ARC-Challenge. Main-manuscript Fig. 4(d) shows that the distillation footprint remains detectable across the evaluated reasoning datasets. Even when evaluated on out-of-domain tasks like MATH (AUROC 0.953) or ARC-Challenge (AUROC 0.912), CML+MTCR distinguishes distilled students from non-distilled controls in this aggregate evaluation. This is cross-dataset operating-envelope evidence, not a broad deployment guarantee. Supplementary Fig. S3(c) extends the same point to the broader cross-task matrix used in the TIFS stress bundle.

H. Sensitivity to Fine-Tuning Hyperparameters

Finally, we probe whether the detection signature is an artifact of specific student training parameters. We sweep the teacher-forcing temperature $T \in \{0.2, 0.5, 0.8, 1.0, 1.2\}$ and the student supervised-fine-tuning learning rate $\eta \in \{5\times10^{-6}, 1\times10^{-5}, 2\times10^{-5}, 5\times10^{-5}\}$. Across all configurations, the detection AUROC remains consistently high (ranging from 0.981 to 0.999), indicating that the capability-matched test is not driven by the particular temperature or learning-rate setting used in the main run. The compact summary is reported in main-manuscript Fig. 4(e,f).